

G1 Probability — Worksheet

IGCSE Mathematics 0580 | Practice Problems

G1_Q1: Basic Probability - Problems

F

Foundation

[1 mark]

Q1. A bag contains 3 red balls and 5 blue balls. What is the probability of picking a red ball?

F

Foundation

[1 mark]

Q2. A fair six-sided dice is rolled once. What is the probability of rolling a 5?

F

Foundation

[1 mark]

Q3. The probability that it rains tomorrow is 0.3. What is the probability that it does NOT rain tomorrow?

F

Foundation

[1 mark]

Q4. A spinner has 4 equal sections coloured red, blue, yellow and green. It is spun once. What is the probability of landing on yellow?

C

Core

[1 mark]

Q5. A box contains 7 red pens, 5 blue pens and 3 black pens. One pen is taken at random. What is the probability that it is blue?

C Core

[1 mark]

Q6. A fair dice is rolled once. Find the probability of rolling a number greater than 4.

C Core

[1 mark]

Q7. A bag contains 2 green marbles, 6 white marbles and 7 purple marbles. One marble is taken at random. What is the probability that it is NOT white?

C Core

[1 mark]

Q8. A card is drawn at random from a standard pack of 52 playing cards. Find the probability that it is a heart or a diamond.

C Core

[1 mark]

Q9. A letter is chosen at random from the word MATHEMATICS. What is the probability that the letter is M?

C Core

[1 mark]

Q10. A bag contains only red, blue and yellow beads. The probability of picking a red bead is 0.2 and the probability of picking a blue bead is 0.5. What is the probability of picking a yellow bead?

E Unknown

[1 mark]

Q11. Two fair dice are rolled and the sum of the numbers is recorded. Find the probability that the sum is 7.

E Unknown

[1 mark]

Q12. Two fair dice are rolled. Find the probability that the product of the two numbers is 12.

E Unknown

[1 mark]

Q13. A bag contains 4 red, 5 white and 6 blue marbles. Three marbles are drawn at random with replacement. Find the probability that at least one marble is red.

E Unknown

[1 mark]

Q14. A spinner has numbers 1 to 8, each equally likely. The spinner is spun twice. Find the probability that both spins land on an even number.

X Extension

[1 mark]

Q15. Three fair dice are rolled. Find the probability that the three numbers form a strictly increasing sequence (e.g. 2, 3, 5 but not 2, 2, 4).

X

Extension

[1 mark]

Q16. A bag contains 5 white and 7 black balls. Two balls are drawn at random without replacement. Find the probability that they are of different colours.

G1_Q2: Sample Space Diagrams - Problems

F

Foundation

[1 mark]

Q17. Two fair coins are tossed. List all possible outcomes in the sample space.

F

Foundation

[1 mark]

Q18. Two fair coins are tossed. What is the probability of getting exactly one head?

F

Foundation

[1 mark]

Q19. A spinner with numbers 1, 2, 3 is spun and a coin is tossed. What is the probability of getting (3, Heads)?

F

Foundation

[1 mark]

Q20. Two fair dice are rolled and their scores are added. How many outcomes give a total of 5?

C Core [1 mark]

Q21. Two fair dice are rolled. Using a sample space diagram, find the probability that both numbers are the same (a double).

C Core [1 mark]

Q22. Two fair dice are rolled. Find the probability that the sum of the two numbers is 8 or more.

C Core [1 mark]

Q23. A fair coin is tossed and a fair six-sided dice is rolled. Find the probability of getting a head and an even number.

C Core [1 mark]

Q24. Two fair dice are rolled. Find the probability that the sum of the two numbers is a prime number.

C Core [1 mark]

Q25. Two fair dice are rolled. Find the probability that the number on the first dice is greater than the number on the second dice.

C Core

[1 mark]

Q26. Two fair dice are rolled. Find the probability that the sum of the two numbers is a multiple of 3.

E Unknown

[1 mark]

Q27. Two fair dice are rolled and the difference between the two scores (larger minus smaller) is recorded. Find the probability that the difference is exactly 3.

E Unknown

[1 mark]

Q28. Two fair dice are rolled. Find the probability that the sum of the two numbers is at most 9.

E Unknown

[1 mark]

Q29. A bag contains red (R) and white (W) cards. Two cards are drawn from the bag. The sample space is {RR, RW, WR, WW} all equally likely. Given that exactly one card is red, find the probability that the first card is red.

E Unknown

[1 mark]

Q30. Two fair dice are rolled. Find the probability that the product of the two numbers is at least 18.

X

Extension

[1 mark]

Q31. Three fair dice are rolled. Find the probability that the sum of the three numbers is 10.

X

Extension

[1 mark]

Q32. Two fair tetrahedral dice (numbered 1 to 4) are rolled, and the two numbers showing on the base are recorded. Find the probability that the sum of the two numbers is a prime number.

G1_Q3: Tree Diagrams - Problems

F

Foundation

[1 mark]

Q33. A coin is tossed twice. Draw a tree diagram and find the probability of getting two heads.

F

Foundation

[1 mark]

Q34. A fair spinner numbered 1 to 4 is spun twice. Find the probability of getting a 3 on the first spin and a 1 on the second spin.

F

Foundation

[1 mark]

Q35. A bag contains 2 red and 3 blue marbles. A marble is taken, replaced, then another is taken. Find the probability of picking two red marbles.

F Foundation [1 mark]

Q36. A fair spinner with colours red (R), blue (B), green (G) is spun twice. Find the probability of getting the same colour both times.

C Core [1 mark]

Q37. The probability that it rains on any given day is 0.4. Using a tree diagram, find the probability that it rains on exactly one of the next two days.

C Core [1 mark]

Q38. A bag contains 3 red and 2 blue sweets. Two sweets are taken without replacement. Draw a tree diagram and find the probability that both sweets are red.

C Core [1 mark]

Q39. A bag contains 3 red and 2 blue sweets. Two sweets are taken without replacement. Find the probability that they are different colours.

C Core [1 mark]

Q40. The probability that a seed germinates is 0.8. Three seeds are planted. Find the probability that all three germinate.

C Core [1 mark]

Q41. In a game, the probability of winning a prize is 0.3. A player plays the game twice. Find the probability that the player wins exactly one prize.

C Core [1 mark]

Q42. A box contains 4 red pens and 6 blue pens. Two pens are taken without replacement. Find the probability that both pens are the same colour.

E Unknown [1 mark]

Q43. A bag contains 5 red, 3 green and 2 yellow marbles. Two marbles are drawn without replacement. Find the probability that both marbles are green.

E Unknown [1 mark]

Q44. A bag contains 5 red, 3 green and 2 yellow marbles. Two marbles are drawn without replacement. Find the probability that the two marbles are of different colours.

E Unknown [1 mark]

Q45. A bag contains 4 white and 6 black balls. Three balls are drawn without replacement. Find the probability that at least one ball is white.

E Unknown

[1 mark]

Q46. A bag contains 3 red and 7 blue marbles. Two marbles are drawn. Find the probability that the second marble is blue, given that the first marble drawn was red and not replaced.

X Extension

[1 mark]

Q47. A box contains 2 red, 3 white and 5 green balls. Three balls are drawn without replacement. Find the probability that the three balls are all different colours.

X Extension

[1 mark]

Q48. In a school, the probability that a student studies Mathematics is 0.6, Physics is 0.5, and Chemistry is 0.4. The choices are independent. Find the probability that a student studies exactly two of these three subjects.

G1_Q4: Conditional Probability - Problems

F Foundation

[1 mark]

Q49. A bag contains 3 red and 5 blue marbles. One marble is drawn and it is red. It is not replaced. What is the probability that the second marble drawn is also red?

F Foundation

[1 mark]

Q50. A fair dice is rolled. Given that the outcome is an even number, what is the probability that it is a 6?

F

Foundation

[1 mark]

Q51. A card is drawn from a standard deck of 52 cards. Given that it is a red card, what is the probability that it is a heart?

F

Foundation

[1 mark]

Q52. In a class of 30 students, 18 are girls and 12 are boys. 10 of the girls wear glasses. Given that a student chosen at random is a girl, what is the probability she wears glasses?

C

Core

[1 mark]

Q53. In a survey of 100 students: 60 study French, 40 study German, 25 study both. A student is chosen at random. Given that the student studies French, what is the probability they also study German?

C

Core

[1 mark]

Q54. A bag contains 4 red and 6 blue marbles. Two marbles are drawn without replacement. What is the probability that the second marble is red, given that the first marble was blue?

C

Core

[1 mark]

Q55. A fair dice is rolled twice. Given that the sum of the two numbers is 7, what is the probability that the first dice shows a 3?

C Core

[1 mark]

Q56. A card is drawn from a standard deck. Given that it is a face card (Jack, Queen or King), what is the probability that it is a King?

C Core

[1 mark]

Q57. A two-way table shows:

	Pass	Fail	Total
Boys	28	12	40
Girls	35	5	40
Total	63	17	80

A student is selected at random. Given that the student passed, find the probability that the student is a girl.

C Core

[1 mark]

Q58. A bag contains 5 red, 3 blue and 2 green marbles. One marble is drawn and it is NOT green. What is the probability that it is red?

E Unknown

[1 mark]

Q59. A bag contains 3 red and 7 black balls. Two balls are drawn without replacement. Given that the balls are of different colours, find the probability that the first ball was red.

E Unknown

[1 mark]

Q60. The probability that a patient has a disease is 0.02. A test gives a positive result for 95% of people with the disease and for 3% of people without the disease. If a person tests positive, find the probability that they actually have the disease.

E Unknown

[1 mark]

Q61. A bag contains 4 red and 6 white balls. Two balls are drawn without replacement. Given that at least one ball is red, find the probability that both balls are red.

E Unknown

[1 mark]

Q62. A fair dice is rolled. Given that the result is a prime number, find the probability that it is also odd.

X Extension

[1 mark]

Q63. A bag contains 6 red and n white balls. Two balls are drawn without replacement. The probability that both balls are red is $\frac{1}{3}$. Find n .

X Extension

[1 mark]

Q64. A factory has two machines A and B. Machine A produces 60% of the items, machine B produces 40%. Machine A makes 2% defective items, machine B makes 5% defective items. An item is chosen at random and found to be defective. Find the probability that it came from machine A.

G1_Q5: Venn Diagrams & Probability - Problems

F Foundation

[1 mark]

Q65. In a Venn diagram, set A has 5 elements and set B has 8 elements. The intersection has 3 elements. How many elements are in $A \cup B$?

F Foundation [1 mark]

Q66. A Venn diagram shows 20 students. 12 play football, 8 play tennis, 4 play both. A student is chosen at random. What is the probability they play neither sport?

F Foundation [1 mark]

Q67. A Venn diagram shows 30 people. 15 like tea, 12 like coffee, 7 like both. What is the probability that a randomly chosen person likes coffee but not tea?

F Foundation [1 mark]

Q68. In a class of 25 students, 18 study Maths and 15 study English. 10 study both. Find the probability that a randomly chosen student studies Maths or English (or both).

C Core [1 mark]

Q69. There are 40 students. 22 study Biology, 18 study Chemistry, 8 study neither. Draw a Venn diagram and find how many study both.

C Core [1 mark]

Q70. In a group of 50 people, 30 like cats, 25 like dogs, and 10 like both. A person is chosen at random. Find the probability that they like exactly one type of pet.

C

Core

[1 mark]

Q71. In a survey of 60 people, 35 read newspaper A, 28 read newspaper B, and 12 read neither. Find the probability that a randomly chosen person reads both newspapers.

C

Core

[1 mark]

Q72. $\xi = \{1,2,3,4,5,6,7,8,9,10\}$. $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7,8\}$. A number is chosen at random from ξ . Find $P(A \cap B)$.

C

Core

[1 mark]

Q73. $\xi = \{1,2,3,4,5,6,7,8,9,10,11,12\}$. $P = \{\text{multiples of 3}\}$, $Q = \{\text{factors of 12}\}$. A number is chosen at random. Find $P(P \cup Q)$.

C

Core

[1 mark]

Q74. In a group of 40 students, 25 play cricket, 20 play football, and 5 play neither. A student is chosen at random. Given that they play cricket, find the probability they also play football.

E

Unknown

[1 mark]

Q75. $\xi = \{1,2,3,\dots,20\}$. $A = \{\text{even numbers}\}$, $B = \{\text{multiples of 3}\}$, $C = \{\text{multiples of 5}\}$. A number is chosen at random. Find $P(A \cap B \cap C)$.

E Unknown

[1 mark]

Q76. In a survey of 80 students, 42 take History, 36 take Geography, and 28 take both. A student is chosen at random. Find the probability that they take exactly one of these subjects.

E Unknown

[1 mark]

Q77. A and B are two events such that $P(A) = 0.6$, $P(B) = 0.4$, and $P(A \cup B) = 0.8$. Find $P(A \cap B)$.

E Unknown

[1 mark]

Q78. Given that $P(A) = 0.5$, $P(B) = 0.3$, and $P(A \cap B) = 0.15$, find $P(A' \cap B')$.

X Extension

[1 mark]

Q79. In a group of 100 students, 55 study French, 45 study German, 30 study Spanish, 20 study French and German, 15 study French and Spanish, 10 study German and Spanish, and 5 study all three. Find the probability that a randomly chosen student studies exactly two languages.

X Extension

[1 mark]

Q80. In a class, $P(M) = 0.7$, $P(P) = 0.6$, $P(M \cap P) = 0.4$, where M = studies Maths, P = studies Physics. Find $P(M' \cap P)$.

G1_Q6: Expected Value - Problems

F

Foundation

[1 mark]

Q81. A fair dice is rolled 60 times. How many times would you expect to roll a 4?

F

Foundation

[1 mark]

Q82. A bag contains 3 red and 5 blue marbles. A marble is drawn, its colour noted, and replaced. If this is done 80 times, how many times would you expect to draw a red marble?

F

Foundation

[1 mark]

Q83. A spinner has numbers 1 to 8, each equally likely. If the spinner is spun 200 times, how many times would you expect to get a 7?

F

Foundation

[1 mark]

Q84. A coin is tossed 300 times. How many times would you expect to get heads?

C

Core

[1 mark]

Q85. In a game, you win 5 with probability 0.3 and lose 2 with probability 0.7. Find the expected value of the game.

C Core

[1 mark]

Q86. A fair dice is rolled 180 times. How many times would you expect to roll a number greater than 4?

C Core

[1 mark]

Q87. A card is drawn from a standard deck and then replaced. If this is done 260 times, how many times would you expect to draw a heart?

C Core

[1 mark]

Q88. A raffle sells 100 tickets at 1 each. There is one prize of 50, two prizes of 10 each, and five prizes of 5 each. Find the expected winnings for a person who buys one ticket.

C Core

[1 mark]

Q89. A bag contains 2 red, 3 white and 5 black marbles. A marble is drawn, its colour noted, and replaced. If this is done 200 times, how many times would you expect to draw a white marble?

C Core

[1 mark]

Q90. A game costs 2 to play. You roll a fair dice. If you roll a 6, you win 6. If you roll a 1, you win 4. Otherwise you win nothing. Find the expected profit (or loss) per game.

E Unknown

[1 mark]

Q91. A game involves spinning a fair spinner numbered 1 to 5. You win the square of the number in dollars. Find the expected winnings per spin.

E Unknown

[1 mark]

Q92. Two fair dice are rolled and the sum of the two numbers is recorded. If this is done 108 times, how many times would you expect the sum to be 5?

E Unknown

[1 mark]

Q93. A lottery ticket costs 5. The probability of winning 100 is 0.02, the probability of winning 20 is 0.05, the probability of winning 10 is 0.1, and otherwise you win nothing. Find the expected profit or loss per ticket.

E Unknown

[1 mark]

Q94. A bag contains 4 red balls and 6 blue balls. You draw a ball at random. If it is red you win 3, if it is blue you lose 2. Find the expected value of this game.

X Extension

[1 mark]

Q95. A game costs c to play. You roll two fair dice. If the sum is 7, you win 10. If the sum is 2 or 12, you win 20. Otherwise you win nothing. Find the value of c that makes the game fair (expected profit = 0).

X Extension [1 mark]

Q96. A game costs 4 to play. You draw two cards from a standard deck without replacement. If both are aces, you win 100. If exactly one is an ace, you win 10. Otherwise you lose. Find the expected profit of this game.

G1_Q7: Experimental vs Theoretical Probability - Problems

F Foundation [1 mark]

Q97. A coin is tossed 50 times and lands on heads 28 times. What is the experimental probability of getting heads?

F Foundation [1 mark]

Q98. A dice is rolled 60 times and a 3 appears 8 times. What is the experimental probability of rolling a 3?

F Foundation [1 mark]

Q99. In a survey of 200 people, 120 said they prefer tea over coffee. What is the experimental probability that a person prefers tea?

F Foundation [1 mark]

Q100. A spinner is spun 40 times and lands on red 15 times. What is the experimental probability of the spinner landing on red?

C Core [1 mark]

Q101. A dice is rolled 200 times. The results are shown:

Number: 1 2 3 4 5 6

Freq: 34 38 30 36 28 34

Find the experimental probability of rolling a 6.

C Core [1 mark]

Q102. A fair dice is rolled 60 times. The theoretical probability of rolling a 4 is $\frac{1}{6}$. If the experimental probability is $\frac{11}{60}$, how many times did a 4 actually appear?

C Core [1 mark]

Q103. In an experiment, a drawing pin lands 'point up' 72 times out of 200 tosses. If the pin is tossed 500 times, estimate how many times it will land 'point up'.

C Core [1 mark]

Q104. A fair coin is tossed 100 times and lands on heads 53 times. What is the difference between the experimental probability and the theoretical probability of getting heads?

C Core [1 mark]

Q105. A bag contains an unknown number of red and blue marbles. In 80 draws with replacement, 32 were red. Estimate the probability of drawing a red marble.

C Core

[1 mark]

Q106. A spinner with 5 equal sections numbered 1 to 5 is spun 500 times. The table shows the results:

Number: 1 2 3 4 5

Frequency: 98 112 95 103 92

Find the experimental probability of spinning an odd number.

E Unknown

[1 mark]

Q107. A fair dice is rolled 300 times. The number 4 appears 42 times. Find the percentage error between the experimental probability and the theoretical probability.

E Unknown

[1 mark]

Q108. A biased coin lands on heads 32 times out of 80 tosses. If the coin is tossed 450 times, estimate the number of heads expected.

E Unknown

[1 mark]

Q109. A spinner is spun many times. The experimental probability of landing on blue is 0.32. If the spinner has 5 equal sections (red, blue, green, yellow, orange), is the spinner likely fair? Explain your reasoning.

E Unknown

[1 mark]

Q110. A dice is rolled 120 times. The experimental probability of rolling a prime number is 0.55. How many prime numbers appeared?

X

Extension

[1 mark]

Q111. A fair coin is tossed n times. After 10 tosses, the experimental probability of heads is 0.7. After 1000 tosses, the experimental probability is 0.501. Explain this change in terms of the Law of Large Numbers.

X

Extension

[1 mark]

Q112. Two dice are rolled 180 times. The sum of 7 appears 28 times. Compare the experimental probability with the theoretical probability. How many additional times would sum 7 need to occur in the next 120 rolls for the experimental probability to equal the theoretical probability?